

Agent Model Specification: Trust Game with Affective Precision

Hidden State Factors

Factor	Symbol	States	Controllable?	Notes
Partner type	s{type}	cooperator, reciprocator, exploiter, random (4)	No	Small stochastic drift p_switch = 0.05
Partner stance	s{stance}	trusting, neutral, hostile (3)	Indirect	Partner's disposition toward agent. Agent influences via actions, cannot observe directly.
Interaction context	s{context}	partner_1, ..., partner_N (N)	Yes (agent_choice mode)	Which partner is currently active
Affective precision	s{beta}	0.2, 0.5, 1.0, 1.5, 2.0 (5)	No	Per-partner metacognitive precision. 1.0 = baseline.

Observation Modalities

Modality	Symbol	Outcomes	Depends on
Partner action	o{action}	cooperate, defect (2)	s{type}, s{stance}

Modality	Symbol	Outcomes	Depends on
Payoff	$o\{\text{payoff}\}$	e.g. $\{-1, 1, 3, 5\}$ (4)	agent_action x partner_action (see note below)
Interoceptive	$o\{\text{intero}\}$	5 levels of affective charge	$s\{\text{beta}\}$ only

Payoff is a deterministic function of (agent_action , partner_action), not a standard hidden-state-to-observation mapping.

Control Factor

Factor	Symbol	Actions	Controls
Social action	$\pi\{\text{social}\}$	cooperate, defect (2)	$s\{\text{stance}\}$ transitions via action-dependent $B[1]$

In agent_choice mode the action space expands to $N \times 2$ (choose partner x choose social action), and $\pi\{\text{social}\}$ also controls $s\{\text{context}\}$.

A Matrices

A[0]: $P(o\{\text{action}\} \mid s\{\text{type}\}, s\{\text{stance}\})$

Shape: (2, 4, 3) – 2 outcomes x 4 types x 3 stances.

Values are $P(\text{cooperate} \mid \text{type}, \text{stance})$. $P(\text{defect}) = 1 - P(\text{cooperate})$.

	trusting	neutral	hostile
cooperator	0.95	0.80	0.55
reciprocator	0.90	0.70	0.30
exploiter	0.70	0.35	0.10
random	0.60	0.50	0.35

Optional noise: $A_{\text{noisy}} = (1 - \text{noise}) * A_{\text{clean}} + \text{noise} * 0.5$

A[1]: Payoff

N/A

A[2]: $P(o\{intero\} \mid s\{beta\})$

Shape: (5, 5) – 5 interoceptive observations x 5 beta levels. Columns sum to 1.

Construction: for beta level **b_1** and bin center **epsilon_s**:

$A[2][s, 1] = \exp(\alpha * (\sigma_0^2 - \epsilon_s^2) * b_1)$

Normalize each column. Parameters: $\alpha=3.0$, $\sigma_0^2=0.25$,
 $\beta_levels=[0.2, 0.5, 1.0, 1.5, 2.0]$, $\epsilon_centers=[0.0, 0.25, 0.5, 0.75, 1.0]$.

Low beta (0.2) produces a nearly uniform distribution. High beta (2.0) concentrates mass on high-valence observations.

B Matrices

B[0]: $P(s\{type, t+1\} \mid s\{type, t\})$

Shape: (4, 4) – uncontrollable, replicated across actions.

$P(\text{stay same type}) = 0.95$

$P(\text{switch to each other type}) = 0.05 / 3 \approx 0.017$

B[1]: $P(s\{stance, t+1\} \mid s\{stance, t\}, \pi\{social\})$

Shape: (3, 3, 2) – action-dependent.

$\pi\{social\} = \text{cooperate (action 0):}$

	from trusting	from neutral	from hostile
to trusting	0.90	0.30	0.05
to neutral	0.10	0.60	0.35
to hostile	0.00	0.10	0.60

$\pi\{social\} = \text{defect (action 1):}$

	from trusting	from neutral	from hostile
to trusting	0.10	0.05	0.02
to neutral	0.50	0.35	0.18
to hostile	0.40	0.60	0.80

B[2]: $P(s\{context, t+1\} \mid s\{context, t\}, \pi\{social\})$

Shape: (N, N, U). Deterministic partner selection in `agent_choice` mode. Uniform 1/N in random assignment mode.

B[3]: $P(s\{\text{beta}, t+1\} \mid s\{\text{beta}, t\})$

Shape: (5, 5) – uncontrollable, tridiagonal with persistence 0.80 and reflecting boundaries.

	from VL	from L	from M	from H	from VH
to VL	0.90	0.10	0.00	0.00	0.00
to L	0.10	0.80	0.10	0.00	0.00
to M	0.00	0.10	0.80	0.10	0.00
to H	0.00	0.00	0.10	0.80	0.10
to VH	0.00	0.00	0.00	0.10	0.90

C Vectors

C[0]: Preferences over $o\{\text{action}\}$

`C[0] = [0.0, 0.0]`

No preference over partner actions.

C[1]: Preferences over $o\{\text{payoff}\}$

`C[1] = log_softmax(payoff_levels / temperature)`

Ascending preference for higher payoffs.

C[2]: Preferences over $o\{\text{intero}\}$

`C[2] = [0.0, 0.25, 0.5, 0.75, 1.0]` (scaled by preference temperature)

Ascending preference for high-precision interoceptive observations.

In the factorized implementation, C[2] does not directly enter EFE for action selection since beta dynamics are action-independent during rollout. The effect of beta on behavior runs through gamma modulation. C[2] is included for a fully joint EFE implementation.

D Vectors

D[0]: Prior over $s\{\text{type}\}$

`D[0] = [0.25, 0.25, 0.25, 0.25]`

D[1]: Prior over $s\{\text{stance}\}$

`D[1] = [0.2, 0.6, 0.2]` (trusting, neutral, hostile)

D[2]: Prior over $s\{\text{context}\}$

$D[2] = [1/N, \dots, 1/N]$

D[3]: Prior over $s\{\text{beta}\}$

$D[3] = [0.05, 0.15, 0.60, 0.15, 0.05]$

Concentrated at $\text{beta} = 1.0$ (baseline precision, index 2).

E Vector

Uniform. No policy restrictions.

Single-step: $E = [0.5, 0.5]$ (cooperate, defect)

For multi-step horizons: enumerate all length- τ sequences over {cooperate, defect}.

Affective Charge and Gamma Modulation

After each trial observation, before next policy selection. Per partner k :

Step 1 - Prediction error:

$\text{eps}_k = 1 - P_{\text{predicted}}(\text{observed_action}_k)$

$P_{\text{predicted}}$ comes from the pre-observation belief over (type, stance).

Step 2 - Affective charge:

$\text{phi}_k = \alpha * (\sigma_0^2 - \text{eps}_k^2)$

With $\alpha=3.0$, $\sigma_0^2=0.25$:

$\text{eps} = 0.0$ (perfect prediction) $\rightarrow \text{phi} = +0.75$

$\text{eps} = 0.5$ (baseline) $\rightarrow \text{phi} = 0.00$

$\text{eps} = 1.0$ (total surprise) $\rightarrow \text{phi} = -2.25$

Step 3 - Interoceptive observation:

$p = \text{sigmoid}(\text{phi}_k)$

$\text{o_intero}_k = \text{floor}(p * 5)$ # clipped to [0, 4]

Step 4 - Bayesian update on beta:

$\text{prior}_k = B[3] @ \text{posterior}_k_{\text{prev}}$

$\text{likelihood}_k = A[2][\text{o_intero}_k, :]$

$\text{posterior}_k = \text{normalize}(\text{likelihood}_k * \text{prior}_k)$

Step 5 - Gamma modulation:

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E[beta_k] = dot(posterior_k, [0.2, 0.5, 1.0, 1.5, 2.0])
gamma_k   = gamma_base * E[beta_k]
q(pi)     = softmax(-gamma_k * G(pi))

```

In agent__choice mode, gamma is keyed to the first partner implicated by the policy.

Multi-Partner Handling

Rather than a joint factor expansion ($4^N \times 3^N \times 5^N$), each partner k maintains separate belief vectors:

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Posterior over (type, stance): shape (4, 3)
Posterior over beta:           shape (5,)

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Only the active partner's beliefs update each round.